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Semantic Plagiarism Detection of Figures in Scholarly Documents: A Conceptual Framework

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***Abstract*— Scholarly full-text documents on machine learning typically include numerous result figures that convey valuable information, such as experimental outcomes, assessments, and comparisons between models. However, research work often carries a great risk of plagiarism. Plagiarism can be textual as well as plagiarism of figures. The existing literature largely explores the plagiarism in the text; that is any degree of similarity between the texts of the scholarly documents, thus ignoring the figures. This study builds on the previous literature and brings new insights by proposing a conceptual framework of a system for detecting plagiarism in result-figures of scholarly documents. This would involve generating semantically enriched summaries specific to result-figures, which will be achieved by extracting relevant information from the figures themselves including the area under the curve (AUC), as well as their associated captions in full-text documents. To accomplish this, this study propose to classify the extracted figures and analyze them by parsing the figure text, legends, and data plots, using a convolutional neural network classification model like ResNet50 that is pre-trained on 1.2 million images from ImageNet. The specialized candidate figure summaries would then be evaluated against the specialized actual figure summaries using Jaccard similarity and edit distance metrics thus catering the challenging task of detecting plagiarism of figures.**

Keywords—Figure based plagiarism, machine learning, specialized summaries, text mining, result figure parsing.

I. Introduction

Plagiarism can be of many different forms; it basically means to copy information and content from someone's work and using it as your own. Many studies have been done previously to detect textual plagiarism of scholarly documents [1]. However, the clever detection of figure-based plagiarism from scholarly documents remains largely unexplored by the extant literature. A recent study [2] randomly selected over 10,000 articles from leading computer science conference proceedings for ten years (2004-2014). Statistics from this study highlighted that more than 70% publications contained at least one figure, almost 43% contained one table, and around 36% contained minimum one table and one figure. Thus, showing the importance of figures in scholarly documents. Mostly the scholarly full-text documents comprise of textual and non-textual content. The non-textual contents such as result figures, graphs, charts and tables contain very useful information. Additionally, researchers use result figures to present their data, perform comparisons of different experiments, summarize the experimental results etc. Therefore, a knowledge gap will remain if result figures from scholarly documents are not made into use for plagiarism detection [3]. Notable work has been carried out to find academic plagiarism where the plagiarism is performed by replacing the words with their synonyms, therefore, copying the idea of the work [4]. Other studies have been done on the basis of both synonyms and antonyms [5]. Few scholars have focused on figure plagiarism detection [6, 7] however; it poses a challenge as comparing figures semantics can be a tricky task. This study would contribute to the existing literature in terms of detecting plagiarism in result-figures from full-text scholarly documents. It proposes image processing and semantic comparison techniques to detect plagiarism. Our work is an extension to our previous work [3] where we extracted result figures from full-text scholarly documents and created specialized summaries enriched with figure semantics, area under the curve and figure related metadata. We advance this work to conceptually create a mechanism to detect semantic plagiarism of result figures from full-text scholarly documents as shown in Fig. 1.

The contribution of the paper is firstly to extract result figures from full text documents and classify them. We then parsed the line graphs from result figures to get the text and plotted data. Moreover, area under the curve (AUC) for each curve is also calculated. On the other hand we detected captions from the full-text scholarly documents associated with those result figures and used Okapi BM25 [8, p. 3] to find similar lines matching with those captions. The parsed textual data, AUC and figure related metadata from the full-text scholarly documents is used to create figure specialized summaries. Secondly we propose a conceptual framework to detect figure plagiarism by using Jaccard Similarity[9] and Edit Distance[10] using the specialized summaries.

The rest of the paper is organized as follows: Section II contains literature review of the previous studies on textual and figure based plagiarism. It discusses the knowledge gap of existing literature. Section III explains the complete conceptual framework, including the figure specialized summary creation and proposing a framework for detecting plagiarism. Finally, section IV concludes the paper.

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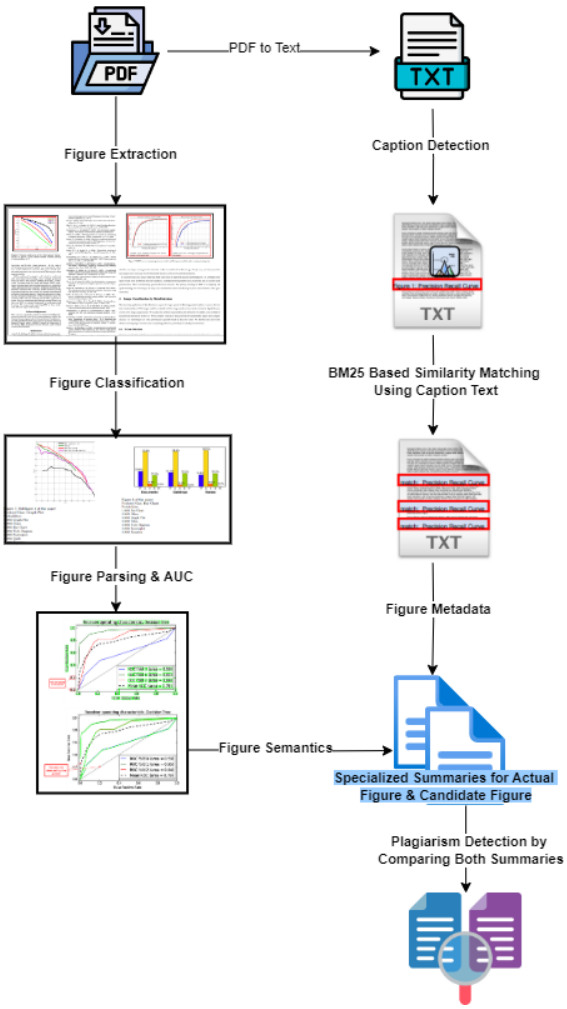

Fig. 1.Proposed conceptual framework for plagiarism detection of figures in scholarly documents.

II. Literature Review

The tremendous increase in academic research poses challenges for researchers and academic writers and in particular, to the journals and information retrieval systems to look for any form of plagiarism. Especially, figure-based plagiarism is a challenging task. To extract information from figures, firstly figures must be localized and extracted from the document. A machine learning based, heuristic independent approach [11] extracts figures from PDF document. In [12] researchers claim about extracting figures from the text irrespective of the format of figure, it uses PDFfigures [13] and has a caption to figure matching component. System also identifies titles, captions and body texts. Another research works [14] put forth an architecture that extracts figures and their metadata such as captions; headings etc., and automatically extracts data from these figures. To make the intended message understood, they then use an NLP module. Different analysis on colored line graph extraction were presented in the form of easy and hard cases, for instance a hard case would be when the curve or the data points are drawn with different colors. Mostly scholarly plots are embedded as vector (SVG). Choudhury et al. [15] argued that information is completely lost if image extraction is done through rasterization of (PNG, JPEG) a PDF page. Whereas, conversion of images to SVG restores all characters in the original image.

Classification of these figures is an essential task. A study [16] claimed that unsupervised classification learning techniques for line graphs performed better than traditional feature learning methods such as SIFT and HoG. Other components of the systems were text extraction from figures and curve extraction. However, the limitation was that it only dealt with colored line graphs. An integrated framework, OverFeat was presented in [17], which trained a convolutional network to simultaneously detect, classify and locate objects in images. A new approach of accumulating bounding boxes rather than suppressing them was used to localize and detect these objects. A number of approaches have been seen to extract information out from these figures. However, very few cater bar charts. [13] automatically extracted text and graphical components from bar charts. To perform this extraction, various image processing as well as heuristics techniques are utilized. [18] presents how semantic structuring of bar charts is done through utilizing machine learning approach. It involves feature extraction from both graphic as well as text components of graphs. After separating text and graphical components, it then determine the roles for extracted components. Hence, feature extraction is done for pixels and for the texts and then classification is performed.

A prominent research in this area is PDFFigures [13] extracts the graphical elements, captions and body text from the full-text documents. Using this approach, pages are decomposed into various parts, for example, graphics, figure texts, body text and captions. It takes a pdf document as input which creates output in the form of tables, graphs, and captions. Scientific literatures presents complex figures, meaning mostly a figure is composition of a number of different subfigures. Separating the multiple subfigures is important to understand these figures. Figures could not be divided into relevant subfigures through PDFFigures. However previously manually designed features and separation rules did this separation task. [19] Uses a data driven approach using deep convolutional neural networks for training of these separators.

ChartSense [20] is a semi-automatic system that extracts data from charts. Firstly, it utilizes a deep learning approach to categorize various types of charts, then an algorithm detects the marks and user interaction does the extraction. A fully automated system known as Scatteract [21] deals with scattered plots with linear scale. It involves utilizing deep learning approach to identify the components of the chart. Then maps the pixel to the chart coordinates with the help of OCR and robust regression. This study focuses on text detection and recognition and data extraction. The system takes in the image of the chart as input, localizes components of the charts such as tick marks and values on Axis and the plots and then it maps the pixel values to the chart coordinates. Manually labelling data of a huge corpus is a strenuous task. Siegel et al. [22] does an automated labelling of a large dataset. We presented an advanced approach in [3] which involves three main steps; first being the extraction of result-figures from scholarly documents, second being the classification of the result-figures through deep learning approach, and lastly parsing those figures in order to extract figure semantics such as text, plotted data and area under the

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curve. Conversely, captions from full-text scholarly document are detected by the system. And system then generates figure specific summaries using the captions from the scholarly document. Finally, it combines the figure semantics from parsing figures and figure metadata from the summaries to generate specialized summaries.

Now the idea is to check for similarity between the extracted figures. AlDabbagh et al. [23] did a comparison of bar charts on the basis of the text extracted from the charts and the numeric values given to the bars. These features include the textual features such as the caption of the figures and the labels of x axis and y axis. And the numeric features which includes the start and end values of the bar charts. The extracted features are then represented in a vector form. It used word 2gram to check similarity between the characters and Euclidean distance to check the similarity between the numeric features. It extracted features from these bar charts which cannot be extracted using OCR tools. [24] Evaluated the similarity of diagrams by analysing both their text and structure. To measure the text similarity, they used a word unigram matching method on diagram components and considered any components with a Jacquard similarity of over 0.5 to be a match. To determine the structural similarity, they calculated the graph edit distance for all matched components. [25] Proposes methods to find similarity in statistical figures such as bar charts and pie charts by extracting silent feature of these figures and storing them as meta data. Then it proposes a strategy to do similarity measure using query formation. [26] Focus on plagiarism detection in flowcharts. It first do some pre-processing on the figures such as removing the connecting lines and texts, then it stores them in database and meanwhile also creates a metadata. It then finds out area of the figures to determine the shape. The plagiarism search engine uses cosine similarity to calculate the level of similarity between the query vector and the figure document. In [27] Method is proposed to find idea and figure plagiarism. It uses image processing techniques to find structural changes, and uses textual features along with the structural features to find the similarities. Moreover, it talks about plagiarism of idea, which we will also try to address in our paper. Another image based plagiarism approached is discussed in [28], where different methods like perceptual and ratio hashing are used against different types of images. [29] Uses features like color, shapes and greyscaling for comparing flowcharts. [30] Uses RGB and HSV color spaces, Tamura texture, and the Canny edge algorithm for feature extraction which are then compared using histogram intersection, euclidean distance, and similarity of edges between the images.

III. Methodology and Proposed Framework

Below is the step by step process for the proposed framework.

Figure Extraction

Documents follow a format for writing the text and presenting figures. Mostly full-text documents follow a similar format, where no blank spaces are left inside the body text, except surrounding the figures. Figures are mentioned with captions. This idea is used for detecting figures from documents for extraction. For Caption identification words like 'Figure: |Fig | Figure |' are looked. Bounding boxes are then used for region identification such as body text, captions, and figures. Each caption is assigned to its relevant figure after the captions and figures are identified.

Occasionally, figures contain another figure within them. For example, a figure on one side can be a graph plot and on the other side, a table representing results from that graph. Therefore, an iterative approach, which involves identification of the split among figures, is used to decompose each figure into subfigures [31].

Figure Classification

Typically, graphs are a frequent form of illustration used to present experimental outcomes. After we have obtained the figures, we proceed to classify them into different types. This is achieved by utilizing a cutting-edge model for figure classification [31]. Our approach involves fine-tuning a pre-trained Convolutional Neural Network (CNN) classifier based on ResNet-50 [32]. The CNN model was initially trained on 1.2 million images from ImageNet [33] and then adapted for the specific task of figure classification.

Figure Parsing

We [3] discussed that parsing figures is a challenging attempt, particularly when it comes to interpreting the results. The figures must adhere to strict specifications, and even slight variations in plot data can significantly impact the final outcome. The complexity is further increased by the fact that figures can be designed in a variety of structures and formatting styles, making it difficult to establish a standard for comparison. Although color can be a useful tool for separating different components within a figure, the same color may be reused, or the figure may be presented in grey scale, making it challenging to differentiate between the plotted parts. Additionally, obstacles such as heavy clutter, distortion, and obscuration can make it difficult to accurately parse the plotted data. We divided figure parsing into the following parts:

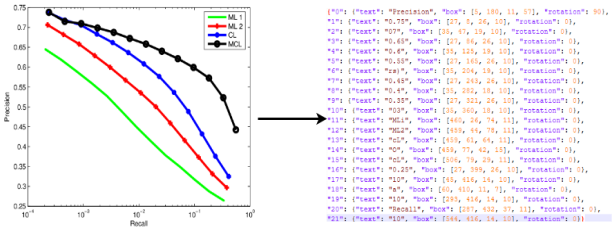

Fig. 2. Jason output for textual parsing

Textual Parsing

We utilized the combination of OpenCV and its deep learning-based OCR tool, Pytesseract (https://pypi.org/project/pytesseract/), for figure parsing. The system extracts the text along with its corresponding bounding box, represented as [x, y, w, h], where x and y are the Cartesian coordinates within the figure, w represents the width, and h represents the height of the bounding box. The width and height are used to determine the orientation of the text, whether it is vertical or horizontal. Through this process, the system extracts text elements such as figure text,

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axis labels, axis scales, and legend text, which are then saved to a JSON file [3]. Fig. 2. Shows the output.

Curve Separation. Parsing figures with multiple curves can be more difficult than those with just a single curve, especially if the curves intersect or if they are displayed in greyscale. To overcome this challenge, we approached the problem by separating the curves based on color. This was achieved by transforming the figure into the HSV color space. Each individual curve was then saved to a separate file. However, we encountered the issue of color variations within a single curve and among different curves. To address this, we analysed the HSV values within and across the curves and set a range for the most common colors. Subsequently, we used Hough transformation to extract points on the curves, which gave us the endpoints of the small lines forming the curves. These endpoints were then sorted in increasing order on the horizontal axis [3]. Fig. 3. shows these steps.

Extracting Area under the Curve. In line graphs, [34] applied the trapezoidal rule to compute the area under the curve. It involved dividing the curve into multiple small intervals in the shape of a trapezoid and computes the area for each small trapezoid. Following is the trapezoidal formula for computing area under the curves with multiple points on the curves:

$$\int_a^b f(x)dx \approx \frac{h}{2}(y_0+y_n + 2(y_1 + y_2 + \cdots + y_{n-1})) \quad (1)$$

Where $y_0 = f(x_0)$, $y1 = f(x_1), y_{n-1} = f(x_{n-1}), y_n = f(x_n)$ and h is the distance between two points on the curve.

We are now able to compare all curves in a figure based on the area and produce some textual description about the figure at hand.

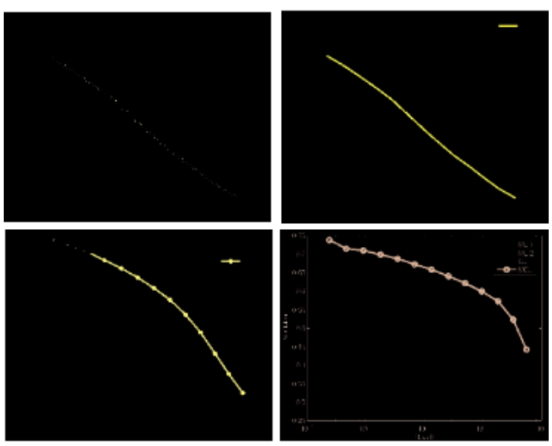

Fig. 3. Curve separation using HSV color space

Summary Generation from Full-Text

A miscellaneous idea of a full-text document is taken from the general summaries. The information about specific document elements might get lost in general summaries. However, a specialized summary focuses on specific elements of a document, tailoring the level of detail to the user's needs. It should provide an appropriate amount of information, ensuring it is neither too long nor too short. Inspired by the work of [35] specialized summaries for the figures are generated in our study. For this the following pre-processing has been done.

PDF to Text

We extracted the plain text from the PDF document by using PDFbox library (https://pdfbox.apache.org/).

Hierarchal Segmentation of Document. A full-text document is composed of different sections such as Abstract, Introduction, Literature review, Methodology, Experiments and Results, Conclusion and References. These sections are organized in a hierarchical manner. The possibility of occurrence of the result figures is higher in Experiments and Results section. Thus, we need to focus on this section and ignore the rest of the paper.

Figure Caption Detection

Generally, figure captions could be found in the figures section, and follow a specific pattern as mentioned below:

<CAPTION> ::= <DOC_EL_TYPE> <integer> <DELIMITER><TEXT>
(2)

The DOC_EL_TYPE here is "Figure," followed by an integer that tracks the specific figure number. This is succeeded by a delimiter, either ":" or ".", and finally, some text that explains the figure.

Average Line Length.

The count of words in a line is known as an average line length. It is computed by summing up the number of words in each line divided by total number of lines. Lines that have length lesser than the average line length are discarded. Following this, we filter out the section headings, titles, etc. As captions are already being detected, hence they are not removed.

Word Density

Sparse lines often result from converting a PDF to text. These lines can be associated to author name, table data, equations etc. These lines are removed using word density measure as follows:

$$dwl = \frac{L}{L+S} \quad (3)$$

dwl = word density of line, L = Length of line, S = No. of spaces in line l.

Similarity to Caption

To generate a specialized summary, we parse the text document and measure the similarity of each line to the caption. Each line is assigned a score, and the top-scoring lines are selected for the summary. For this specialized summary generation we adapt Okapi BM25 [8, p. 3] similarity matching measure.

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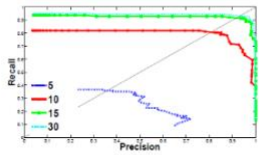

Figure 7. Orientation accuracy. Detection performances for different values of orientation estimation.

System generated Summary

```
Figure 7: Orientation accuracy. Detection performances for
different values of orientation estimation.Another experiment to
measure orientation estimation ...accuracy is shown in Figure 7
For this experiment a true ...Figure 7 Orientation accuracy
Detection performances for differ  ...dataset are shown in
Figure 5 a Spread LBF outperforms ...is below a given accuracy
value The figure shows differ  ...
precision versus recall area under the curve greenCombo perform
better roi =0.96
```

Fig. 4. System generated figure specialized summary

$$BM25(C, L) = \Sigma\{\log\frac{N}{Lf_t} \times \frac{(k_1+1)tf_{tl}}{k_1\left((1-b)+b\times\left(\frac{l_L}{l_{av}}\right)\right)+tf_{tl}} \times \frac{(k_3+1)tf_{tc}}{k_3+tf_{tc}}\} \quad (4)$$

N = No. of lines in the document, L_{f_t} = line frequency, tf_{tl}= frequency of term t in line L, tf_{tc}= frequency of term t in captionC, l_L= length of lineL, l_{av}= average length of line in Document, and the k_1,$k_3$and b are set to 2, 2 and 0.75.

After the textual summary generation, we combined it with figure semantics that we got as a result of figure parsing with the metadata from our figure based summaries from full-text. This gave us figure specialized summaries as shown in Fig. 4.

Plagiarism Detection from Specialized Summaries

The first step is to generate specialized summaries for both the candidate figure and the actual figure. These summaries should capture the key information and visual elements of the figures, such as the axes, labels, legends, and data points. The specialized summaries also include a description of the Area Under the Curve which is a vital finding to be compared and figure related information found in the document.

Once the specialized summaries have been generated, the next step is to compare them to determine the degree of similarity between the candidate figure and the actual figure. We propose the following two approaches or this.

Firstly Jaccard similarity metrics [9] to compare both the summaries i.e. the specialized summary of the candidate figure with that of actual figure.

$$Sim(Sc, Sa) = \left| S_c \cap S_a \right| \div \left| S_c \cup S_a \right| \quad (5)$$

Where Sc is the specialized summary of the candidate figure, Sa is the specialized summary of the actual figure.

The range for Jaccard similarity metrics is from 0 to 1, with values closer to 1 indicating higher similarity between the two summaries. We will associate a value higher than 0.5 to be considered as a plagiarism.

Secondly we propose Edit distance[10] to find dissimilarity between the two summaries. We will calculate the number of operations performed to convert the candidate summary to the actual summary. The larger value of edit distance measure shows greater dissimilarity between the two.

Lastly, we will compare the Jaccard similarity value with the Edit distance. Summaries with a higher Jaccard similarity value should have a smaller Edit distance to ensure plagiarism.

IV. Concluding Remarks

Overall, by comparing the specialized summaries of the candidate figure with the specialized summaries of actual figure, we can determine whether the candidate figure is similar enough to the actual figure to be considered a good match. This process is crucial for evaluating the quality and accuracy of the match of the figures, particularly in scientific research where figures play a critical role in conveying information and findings to the audience.

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